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Project Plan

Graduation Internship

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Table of Contents

1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	31-08-2026	Draft Version	Casey Stijlaart	Draft version	Initial document structure, project assignment and context
0.2	01-09-2026	Updated	Casey Stijlaart	Draft version	Updated assignment and context, approach, project organization, and risk analysis.
0.3	02-09-2026	Final Draft Version	Casey Stijlaart	Final draft version	Grammar and spelling revision, and revision communication and timeline section.

2. Project Overview

2.1. Client

The client for this project is MindLabs, a creative technology studio affiliated with Fontys in Tilburg.

2.2. Virtual Human

A Virtual Human is a real-time, computer-generated character with a human-like appearance and behaviour. It is rendered through a game engine (Unreal Engine, in the case of the Holobox) rather than pre-recorded video, a screen-based avatar, or a physical robot. It combines a 3D model, an animation and rendering pipeline, and interaction logic that lets it respond to a visitor in real time, for example through speech, gaze, gestures, or posture.

For the Holobox's Virtual Human to respond convincingly, for example by kneeling down to a child's eye level, it depends on external input describing where a visitor is and how tall they are. Supplying that positional and height data is exactly what this internship's sensing system is for. The Virtual Human's own appearance, animation, and dialogue remain the responsibility of the Experience Design intern (see Project Organization).

2.3. Motivation

In the context of the Holobox project, a holographic display installation where a Virtual Human interacts with visitors, there is a need for a sensing system that can determine a person's 3D position in real time. This is particularly important to ensure a proper interaction and engagement with the Virtual Human.

2.4. Target Audience

The primary target audience for this project is children aged 8-12, who will interact with the Virtual Human in the Holobox installation.

2.5. TRL (Technology Readiness Level)

The project currently stands at TRL 0, as no work has yet been started on this aspect. As the project progresses, it aims to advance towards higher TRL levels, such as TRL 5/6, through iterative refinement and testing.

3. Project Assignment

3.1. Context

MindLabs is a creative technology studio, affiliated with Fontys, in Tilburg. They develop immersive, interactive experiences at the intersection of the digital and physical world. One of its current projects is the Holobox: a holographic display installation in which a Virtual Human interacts with visitors.

For the Virtual Human to feel truly responsive and natural, it needs to know where a person is in the space, three-dimensionally, so it can adapt its behaviour accordingly, for example by kneeling down when interacting with a child. Currently, no such sensing system exists. A camera-based solution would be an obvious approach, but it raises privacy concerns in a context involving children, and it would offload significant processing to external systems.

There is therefore an opportunity to design a dedicated embedded sensing system that can determine a person's 3D position in real time in a privacy-preserving way, and output that data reliably to the Holobox pipeline. The system also needs to be designed with future extensibility in mind, since scenarios involving multiple people in the space are anticipated.

3.2. Project Goal

The goal of this internship is to research, design, and build an embedded sensing system that estimates the real-time 3D position, including height, of a person inside the Holobox interaction space, without using a camera, so that the Holobox's Virtual Human can adapt its behaviour (e.g. kneeling down for a child) based on that estimate.

Sub-goals

- Identify and compare sensing technologies against criteria relevant to the Holobox environment: spatial resolution, height discrimination, embedded feasibility, and robustness in a real environment with children.
- Select the sensor(s) and MCU/SBC and design the architecture.
- Implement a real-time processing pipeline in C/C++ and/or Python on an embedded platform that turns sensor data into a position and height-category estimate.
- Explore lightweight, on-device machine learning for height classification or person selection, where this improves accuracy or robustness over a purely rule-based approach.
- Document the resulting architecture, data flow, and design decisions so the solution can be understood, maintained, and extended by MindLabs after the internship ends.

The system is designed with future multi-person extensibility in mind, but single-person tracking within the Holobox interaction space is the scope of this internship.

3.3. Assignment

The goal of this internship is to research, design, and build an embedded sensing system capable of estimating the real-time 3D position, including height, of a person within the Holobox interaction space, without the use of cameras.

Key features of the system include:

- Real-time estimation of a single person’s 3D position and height category (adult or child) within the Holobox interaction space.
- ML-based or rule-based non-camera sensor(s) (e.g. mmWave radar, ultrasonic, IR, UWB) to achieve robust position and height estimation under real-world noise conditions.
- An output interface that exposes the position and height estimate to the Holobox’s Unreal Engine pipeline, so the Virtual Human can adapt its behaviour (e.g. kneeling for a child) based on the estimate.
- Documentation of the system architecture, data flow, and design decisions to ensure maintainability and extensibility for future multi-person tracking scenarios.

The final deliverable will be a working embedded prototype that demonstrates the system’s capabilities, along with the supporting documentation, including a technical research report covering sensor selection, implemented interaction logic (position + height output), and a system architecture document. The system will be designed with multi-person extensibility as a future-facing requirement, though single-person tracking is the focus of this internship.

3.4. Scope and Preconditions

Table 2. Scope and Preconditions

The project includes	The project does not include
Test report, to verify the behaviour of the system	The Dutch version of documents
System Design Document	A fully finished prototype ready for a non-controlled environment, as the project is exploratory in nature.
Firmware in the given programming language	
Technical Research Report	

Preconditions Ð during the project multiple technologies will be used to get to the final deliverable. These technologies can include Docker, Visual Studio Code, and GitHub. Besides these technologies, others will be used to develop the firmware, such as C++, along with the necessary hardware.

3.5. Finished Products

At the end of the project, the deliverables consist of documentation and a working prototype, structured as shown in Figure 1.

Figure 1. Project deliverables structure

3.6. Research Questions

3.6.1. Main Research Question

How can the real-time 3D position and height of a single person inside the Holobox interaction space be reliably estimated using non-camera sensor(s), embedded within the Holobox hardware, so that the Virtual Human can respond appropriately?

3.6.2. Sub-questions

Technical sub-questions:

1. What non-camera sensors provides sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?
2. How can sensors be implemented on an embedded platform to produce robust, low-latency position estimates under real-world noise conditions?
3. How can a lightweight ML model be trained and deployed on an edge device to classify person height from sensor data with acceptable accuracy?
4. How should the system architecture be designed to support single-person tracking now while remaining extensible to multi-person scenarios?
5. How should the position and height output be structured and exposed so it can be consumed by the Unreal Engine pipeline with acceptable latency and reliability?

Ethical and legal sub-questions:

1. What is the minimal set of sensor data needed to estimate position and height, and how can the system avoid capturing or storing personally identifiable information, given that the primary visitors are children aged 8-12?
2. What privacy and AVG/GDPR obligations apply to a system that infers physical characteristics (position, height) of children in a public installation, and how should sensor data handling, retention, and processing be designed to comply with them?

User/Human Interaction sub-questions:

1. What position and height estimation latency is required for the Virtual Human’s reaction (e.g. kneeling down for a child) to be perceived by visitors as natural and immediate, rather than delayed or robotic?
2. How should misclassifications (e.g. a child briefly classified as an adult) be handled so they do not produce an interaction that feels wrong or off-putting to the visitor?

Technical sub-questions 1 to 4 are the core research questions identified in the graduation proposal; technical sub-question 5, and the ethical/legal and user/interaction sub-questions, are additional questions derived from the context and assignment scope, and are investigated using the DOT-framework research strategies described in Research Methods.

3.7. Research Methods

In this project, the DOT-framework is used to structure the research and development process. The DOT-framework is a structured approach that focuses on various research strategies and methods to build a system. The research strategies are field, lab, library, workshop, and showroom, as shown in Figure 2.

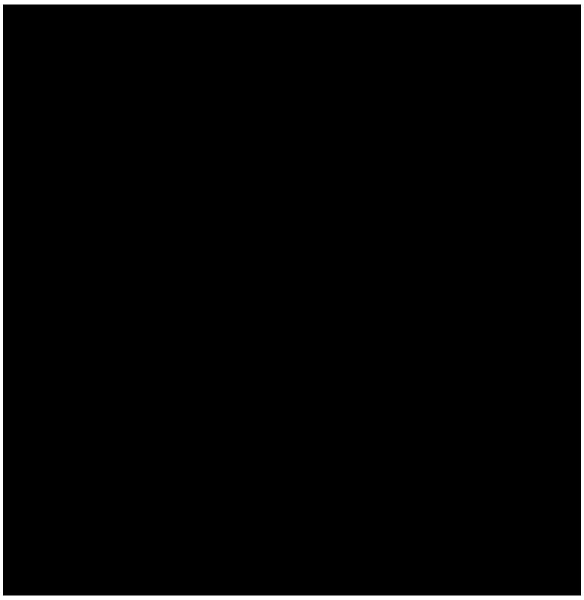

Figure 2. The DOT-framework

3.7.1. Research Methods x Research Questions

Each research question is met with a corresponding research strategy and technique based on the DOT-framework (see Figure 3). This ensures that every question is answered effectively and efficiently, while providing a clear structure for the research process.

Figure 3. Research strategies and techniques

For a comprehensive view of how the DOT-framework strategies align with each research question (see Research Questions), consult Table 3.

Table 3. Research Methods x Research Questions

Research Question	Research Strategy	Research Technique
What non-camera sensors provide sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?	Library, Lab	Available product analysis, literature study, document analysis, and bench experimentation.
How can sensors be implemented on an embedded platform to produce robust, low-latency position estimates under real-world noise conditions?	Lab, Field	Prototyping, benchmarking, and in-situ testing under real-world noise conditions.
How can a lightweight ML model be trained and deployed on an edge device to classify person height from sensor data with acceptable accuracy?	Library, Lab	Literature review, prototyping, model training, and model evaluation.
How should the system architecture be designed to support single-person tracking now while remaining extensible to multi-person scenarios?	Library, Workshop	Design pattern research, IT-architecture sketching, and a design review with the company mentor.
How should the position and height output be structured and exposed so it can be consumed by the Unreal Engine pipeline with acceptable latency and reliability?	Library, Workshop, Showroom	Document analysis of the existing Unreal Engine pipeline, an interface design workshop, and validation through sprint-review demonstrations.

Research Question	Research Strategy	Research Technique
What is the minimal set of sensor data needed to estimate position and height, and how can the system avoid capturing or storing personally identifiable information, given that the primary visitors are children aged 8-12?	Library, Lab, Workshop	Literature study on data minimisation, prototyping to test whether a minimal data set remains accurate enough, and a workshop check with the company mentor on acceptable data use.
What privacy and AVG/GDPR obligations apply to a system that infers physical characteristics (position, height) of children in a public installation, and how should sensor data handling, retention, and processing be designed to comply with them?	Library, Workshop	Literature and document study of AVG/GDPR legislation, and an expert consultation to validate the intended data-handling approach.
What position and height estimation latency is required for the Virtual Human’s reaction (e.g. kneeling down for a child) to be perceived by visitors as natural and immediate, rather than delayed or robotic?	Lab, Field, Showroom	Controlled latency experiments, testing with real users, and stakeholder feedback at sprint-review demonstrations.
How should misclassifications (e.g. a child briefly classified as an adult) be handled so they do not produce an interaction that feels wrong or off-putting to the visitor?	Library, Workshop, Showroom	Literature study on interaction error-handling patterns, prototyping of fallback behaviour, and stakeholder feedback at sprint-review demonstrations.

A structured research log is maintained throughout the project to track how each sub-question is being answered, and findings are validated in the workshop and showroom moments with the company mentor and the Experience Design intern (see Project Organization).

4. Approach and Planning

4.1. Approach

Throughout the project, an agile approach is used to iteratively develop the sensing system, which allows for flexibility and adaptability as the project progresses. Sprints are used, which are short development cycles that allow for rapid prototyping and testing of the system. Each sprint will be 2 weeks long, with a review and retrospective at the end of each sprint to assess progress and identify areas for improvement. This approach is particularly suitable for the exploratory nature of the sensor research, as it allows for rapid prototyping, testing, and refinement of the system based on feedback and findings, as depicted in Figure 4.

Figure 4. Agile methodology

4.1.1. Test Approach

To ensure seamless software implementation, testing has been integrated into the development process from the start. Testing takes on different forms, each tailored to the specific stage of development and the nature of the component being tested. The primary testing methods employed in this project are unit testing, integration testing, and system testing. These methods are designed to validate the functionality, performance, and reliability of the sensing system at various levels of granularity.

4.2. Project Breakdown

The project follows the agile methodology, which allows for iterative development and flexibility in adapting to new findings and requirements. The project is divided into several phases, each with specific objectives and deliverables. These phases include research, design, implementation, testing, and deployment.

1. Requirements & Concept: Define project goal, functional and non-functional requirements, and create a concept for the sensing system.
2. System Design: Plan the overall architecture and select the sensor(s) to use.
3. Sprint Planning: Select tasks from the backlog for the upcoming sprint, define sprint goals, and allocate resources.
4. Implementation: Develop the firmware and software modules incrementally, integrating with the selected hardware.
5. Testing & Integration: Perform unit tests, integration tests, and hardware-in-the-loop tests to ensure the system meets the requirements and functions correctly.
6. Review & Retrospective: Demonstrate the working increment, gather feedback, and reflect on the sprint to identify improvements for the next iteration.

This iterative approach allows for flexibility, early detection of issues, and adaptation based on feedback, alongside the continuous delivery of working increments.

4.3. Timeline

The project’s overall timeline is depicted in Figure 5. This timeline offers a high-level overview of the project phases and significant events.

Figure 5. Project timeline

5. Project Organization

5.1. Team Members

Table 4. Team Members

Role	Name	Organisation	Responsibility
Student / intern	C. Stijlaart	Fontys ICT	Research, design, and implementation of the embedded sensing system
Company mentor	A. Groeneweg	MindLabs (Coordinator and Senior Lecturer UX & Interaction Design)	Weekly guidance on how the sensing system fits into the Holobox experience, interaction requirements, and professional development; contact: a.groeneweg@fontys.nl
Experience Design intern	R. Lengton	MindLabs	Design and behaviour of the Virtual Human.
Fontys examiner / assessor	G. Elbers	Fontys ICT	Academic supervision and assessment of the internship
Fontys 2nd examiner / assessor	M. Beks	Fontys ICT	2nd academic supervision and assessment of the internship

Additional MindLabs team members with technical backgrounds in embedded systems, sensor integration, and software architecture are available to support the technical depth of the project alongside the company mentor's interaction-design guidance.

5.2. Communication

As the project is in a work environment, rather than a traditional academic setting, communication is a key aspect of the project. Rather than having fixed moments for communication, the project is structured around agile sprints, with regular feedback loops and alignment sessions to ensure that the project stays on track and meets the requirements of both the company and the academic institution. There is more flexibility as the environment is focussed on continuous collaboration and communication, rather than a rigid schedule of meetings and reports.

With the assesor from Fontys there is a set schedule. Every other week a meeting will be helped to discuss the process and project status. Furthermore, this is the moment to discuss any questions. For any urgent matter, the assesor can be reached via email or teams.

5.3. Test Environment

Throughout the project, multiple methods are used to test and validate the created software and product. This project’s three most important testing methods are unit testing, integration testing, and static code analysis.

5.3.1. Integration Testing

Integration testing is used to verify that the different components of the system work together as expected.

5.3.2. Unit Testing

Unit testing is used to verify that individual components of the system work as expected. These tests are typically automated and run frequently during development to catch issues early.

5.3.3. Static Code Analysis

Static code analysis is used to analyse the source code for potential issues, such as coding standard violations, security vulnerabilities, and other quality concerns.

5.4. Configuration Management

To ensure the source code is properly managed, a version control system is used. The version control system used is Git, and the repository is hosted on GitHub.

For task management, a GitHub Project board is used to manage the backlog and track the progress of the project. Issues are formulated as user stories, and organised using milestones and labels to keep the backlog structured and the current sprint’s scope visible.

To ensure safety and fallback, work is organised with the help of branches. Each semester of the internship has its own long-lived branch (e.g. `semester-8`), which holds that semester’s stable, up-to-date code. Each task is worked on in a separate branch, which is then merged into the current semester branch once the task is completed.

6. Risk Analysis

6.1. Finance and Risks

6.1.1. Finance

No project budget has been agreed with MindLabs at this stage. Which sensor(s) and embedded hardware are actually needed, and what they would cost, is itself one of the outcomes of the research & sensor selection; it is not something that can be fixed up front. A realistic cost estimate can only be drawn up once the candidate sensors have been narrowed down through the comparison matrix.

Planned approach

- During the research sprints, note the approximate market price of each candidate sensor and embedded platform alongside its technical score in the comparison matrix, so cost is one of the factors in the sensor selection decision.
- Present the shortlisted options and their estimated cost to the company mentor at the Sprint 2 milestone review, together with the recommended sensor(s).
- Only order hardware once the company mentor has approved the specific items and their cost; no purchases are made before that approval, and no budget ceiling is assumed in advance.

Financial risks

- If none of the sensor options the company mentor is willing to fund perform well enough on their own, that cost constraint feeds back into the comparison matrix, resulting in a lower-cost sensor selection.

6.1.2. Risks and Fallback Activities

The main risks identified for this project, along with their likelihood, impact, and fallback activity, are listed in Table 5.

Table 5. Risks and Fallback Activities

Risk	Likelihood	Impact	Fallback activity
A single non-camera sensor does not reliably discriminate height across the full range (children aged 8-12 and adults)	Medium	High	Combine it with one or more complementary sensors (fusion); if fusion is insufficient, narrow scope to coarse height categories (e.g. child / adult) instead of continuous height.
Sensor pipeline does not meet real-time latency requirements on the embedded platform	Medium	Medium	Move processing stages to a more capable platform (e.g. from MCU to Linux SBC), or simplify the processing pipeline/reduce on-device ML complexity.
On-device machine learning underperforms, or the week 12 deployment milestone is not met	Medium	Medium	Fall back to a rule-based/statistical classification approach for height category or person selection instead of ML, and report the deviation against the planning in the mentoring session.
Selected hardware (sensor modules, embedded platform) has long lead times, is unavailable, or costs more than the company mentor is willing to approve	Low	Medium	Identify hardware needs and get budget approval as early as possible during the research sprints (see Finance, above); keep a second, cheaper candidate sensor in the comparison matrix as a fallback option.

Risk	Likelihood	Impact	Fallback activity
Interaction requirements from the Experience Design intern change or are unclear, affecting the system's output requirements	Medium	Medium	Rely on frequent alignment sessions (see Project Organization) to catch requirement changes early, and keep the output simple and versioned so it can absorb small changes.
Time pressure due to the exploratory nature of sensor research delays implementation	Medium	High	Timebox the research & sensor selection sprints to weeks 1-4 (see Approach and Planning); if a decision cannot be made in time, select the best-performing option found so far and proceed.
Testing with real child participants (aged 8-12) in the Holobox is limited or restricted	High	Medium	Use adult testers, and rely on the mock-up test environment described in Project Organization for controlled measurements.